For the Game Major Students

Course Orientation

Game Development Department
International College, DSU
Fall 2024

Dongseo University
동서대학교

Curriculum for Game Development Department

	Major Foundation			Major Application			Major Advanced	
	1-1	1-2	2-1	2-2	3-1	3-2	4-1	4-2
Creativity Competence		Fundamentals of Game Programming	Game Design				Research Methods	Portfolio
Practical Application Competence			Game Algorithm	Database System	Computer Network	Operating System		
			Linear Algebra	2D Game Programming	Programming Method	3D Game Programming		
				Game Tool Programming	Computer Graphics Programming			
				Computer Architecture	Artificial Intelligence Programming			
Convergence Competence							Game Project	
						Game Server Programming	Bachelor Thesis 1	Bachelor Thesis 2
							Using Engine Game	Sound Workshop

Career Path

- 2D/3D Game Client Developer
- Game AI Developer
- Game server Developer

- Design Convergence Developer
- Video Convergence Developer
- XR Developer
- AI Convergence Developer
- eLearning Developer
- Architectural Convergence Developer

	Major Foundation		Major Application			Major Advanced		
	1-2	2-1	2-2	3-1	3-2	4-1	4-2	
Common Subjects	Fundamentals of Game Programming						Portfolio	
Creativity Competence Practical Application Competence		Game Design	2D Game Programming	Programming Method		Game Project		2D Client MD
Convergence Competence			Game Tool Programming	Computer Graphics Programming	3D Game Programming	Using Engine Game		3D Client MD

	Major Foundation		Major Application			Major Advanced		
	1-2	2-1	2-2	3-1	3-2	4-1	4-2	
Common Subjects	Fundamentals of Game Programming						Portfolio	
Practical Application Competence Convergence Competence		Linear Algebra	Computer Architecture	Artificial Intelligence Programming	Operating System			Game AI MD

	Major Foundation		Major Application			Major Advanced		
	1-2	2-1	2-2	3-1	3-2	4-1	4-2	
Common Subjects	Fundamentals of Game Programming						Portfolio	
Practical Application Competence Convergence Competence		Game Algorithm	Database System	Computer Network	Game Server Programming			Game Server MD

Agreement between MRU and DSU

DUAL DEGREE PARTNERSHIP AGREEMENT CONCERNING THE IMPLEMENTATION OF THE JOINT STUDY PROGRAMME IN INFORMATICS AND DIGITAL CONTENTS

No. 2A-45

Mykolas Romeris University (Lithuania) hereinafter referred to as MRU, represented by Rector of the MRU Prof. Dr. Inga Žalėnienė, and Dongseo University (South Korea) hereinafter referred to as DSU, represented by President of the DSU Dr. Jekuk Chang, hereinafter jointly referred to as the Universities or individually referred to as a University, hereby agree to renew the Dual Degree Partnership Agreement Concerning the Implementation of the Joint Study Programme in Informatics and Digital Contents Nr. 2A-17 signed between the Universities on November 11th, 2013 and establish the following Dual degree partnership agreement concerning the implementation of the joint study programme in Informatics and Digital Contents, hereinafter referred to as Agreement.

1. General Provisions – Scope and Purpose of the Agreement

1.1. This Agreement is concluded in order to establish general principles and terms of cooperation between the Universities for implementation of undergraduate joint study programme in Informatics and Digital Contents, hereinafter referred to as the Programme. The titles of the Programme in National languages of the Universities are – Informatika ir skaitmeninis turinys (in Lithuanian) and 디지털콘텐츠 (in Korean). Both Universities have a right to offer undergraduate study programmes and to confer bachelor's degrees according to their respective national legislation.

1.2. The Programme shall be implemented in compliance with the binding national laws, internal

	5 semester		
1	Elective subject 1 (Game or Animation)	DSU	6
2	Programming Method	DSU	6
3	Computer Graphics Programming	DSU	6
4	Artificial Intelligence Programming	DSU	6
5	Computer Network	DSU	6
	6 semester		
1	Elective subject 2 (Game or Animation)	DSU	4
2	3D Game Programming	DSU	6
3	Game Server Programming	DSU	6
4	Game Tool Programming	DSU	6
5	Computer Architecture	DSU	6
6	Operating System	DSU	6
	7 semester		
1	Bachelor Thesis 1	DSU	8
2	Research Methods	DSU	6
3	Using Engine Game	DSU	6
4	Game Project	DSU	6
	8 semester		
1	Bachelor Thesis 2	DSU	8
2	Portfolio	DSU	16
3	Sound Workshop	DSU	6

Programming Method – Syllabus

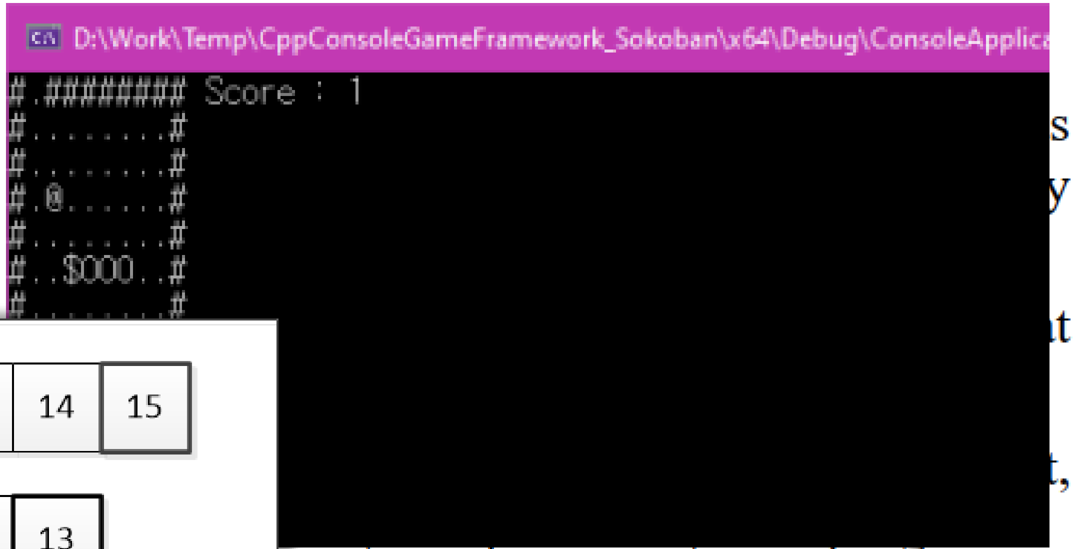
Course Description

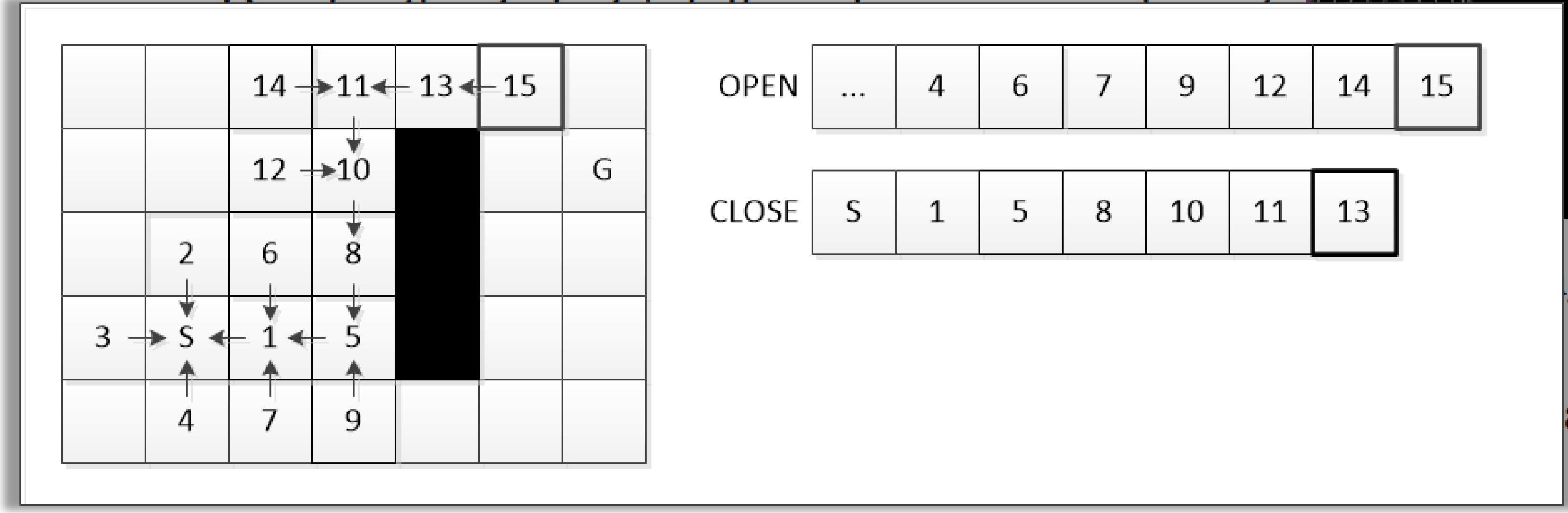
C is a general-purpose programming language with features economy of expression, modern flow control and data structures, and a rich set of operators. We learn basic programming skills with C.

Goals

The main goals of this course are to:

- Introduce the students to the basic concepts and pri... s data types, operators, expressions, control structure... y management.





...t

...t,

...g, such as data structures, algorithms,

...ations of C programming in game

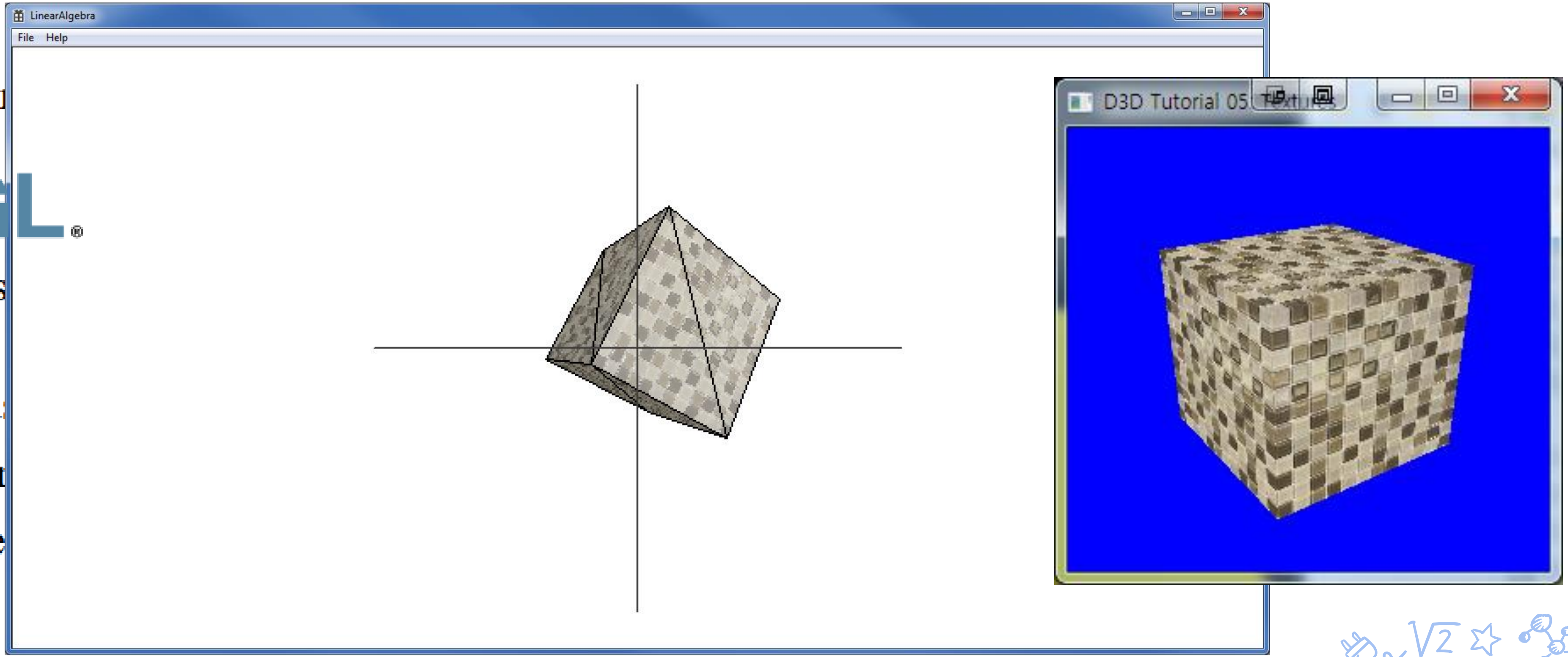
Computer Graphics Programming – Syllabus

Course Description

The aim of this class is to understand the technical principles of color representation implemented in computers and to express various images by applying each unique expression method through computer graphics programs. Students will learn the principles and functions of computer graphics and practices computer graphics techniques using python based programming to visually express creative ideas.

The main goals of this cou

- U erstand the b
- Develop the skills
- Learn how to use
  implement variou
- Explore the curren
- Apply the knowle



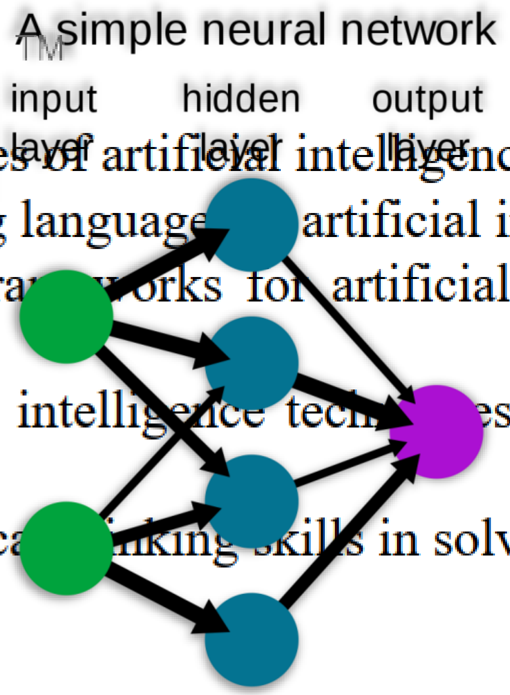
Artificial Intelligence Programming – Syllabus

Course Description

Artificial intelligence (AI) is intelligence demonstrated by machines. Example tasks in which this is done include speech recognition, computer vision, and translation between (natural) languages. We learn programming techniques for general purposes or game AI.

Goals

The main goals of this course are to:



- Understand the basic concepts and principles of artificial intelligence and its applications.
- Know how to use Python as a programming language for artificial intelligence tasks.
- Gain familiarity with various tools and frameworks for artificial intelligence development, such as NumPy, TensorFlow, and mlagent.
- Implement and evaluate different artificial intelligence techniques, such as neural networks, gradient descent, and cross-entropy.
- Develop and demonstrate creative and critical thinking skills in solving artificial intelligence problems.

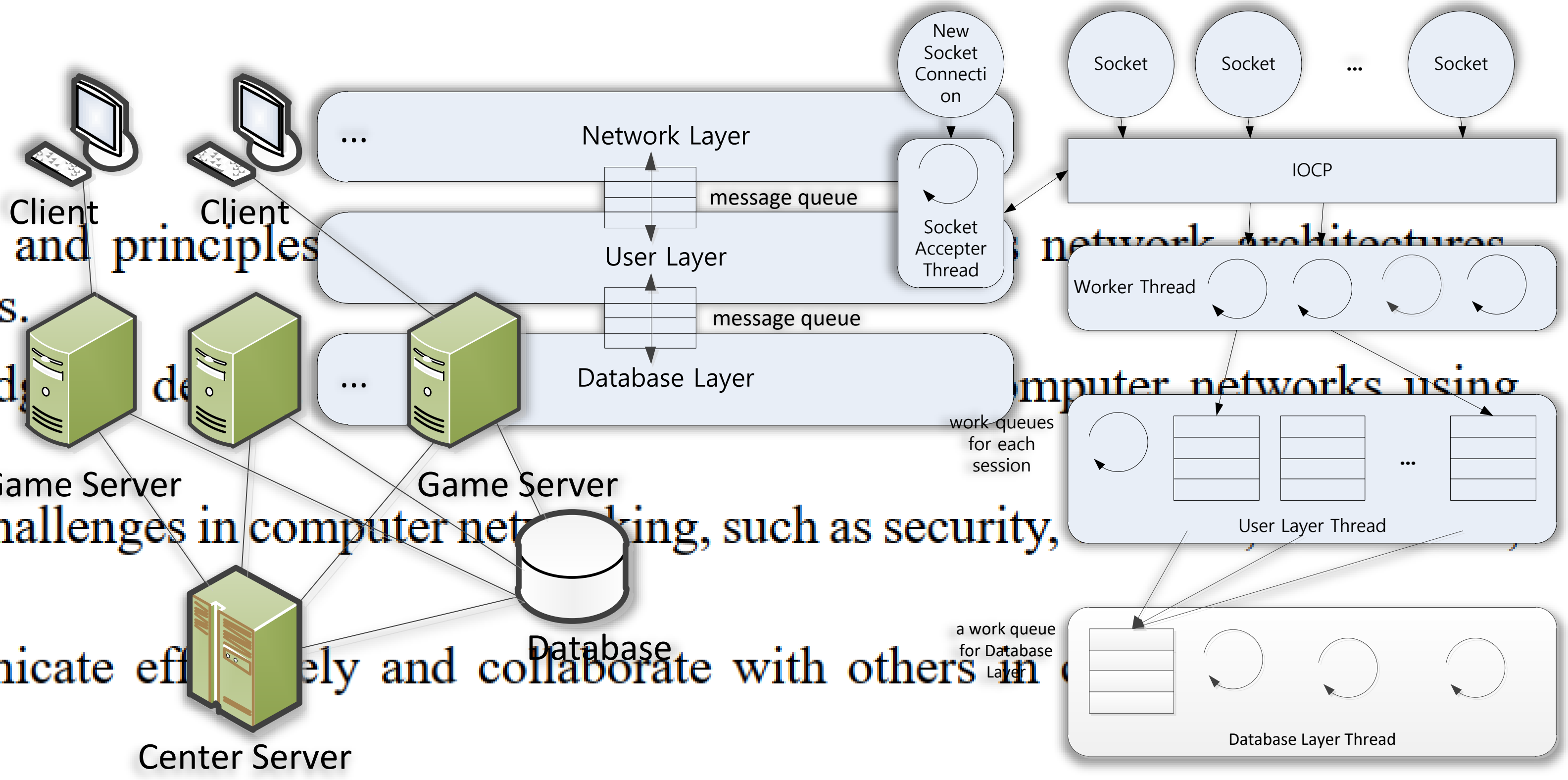
Computer Network – Syllabus

Course Description

A computer network is a digital telecommunications network for sharing resources between nodes, which are computing devices that use a common telecommunications technology. We can summarize all the network components for any type of network.

Goals

The main goals of this course are to:



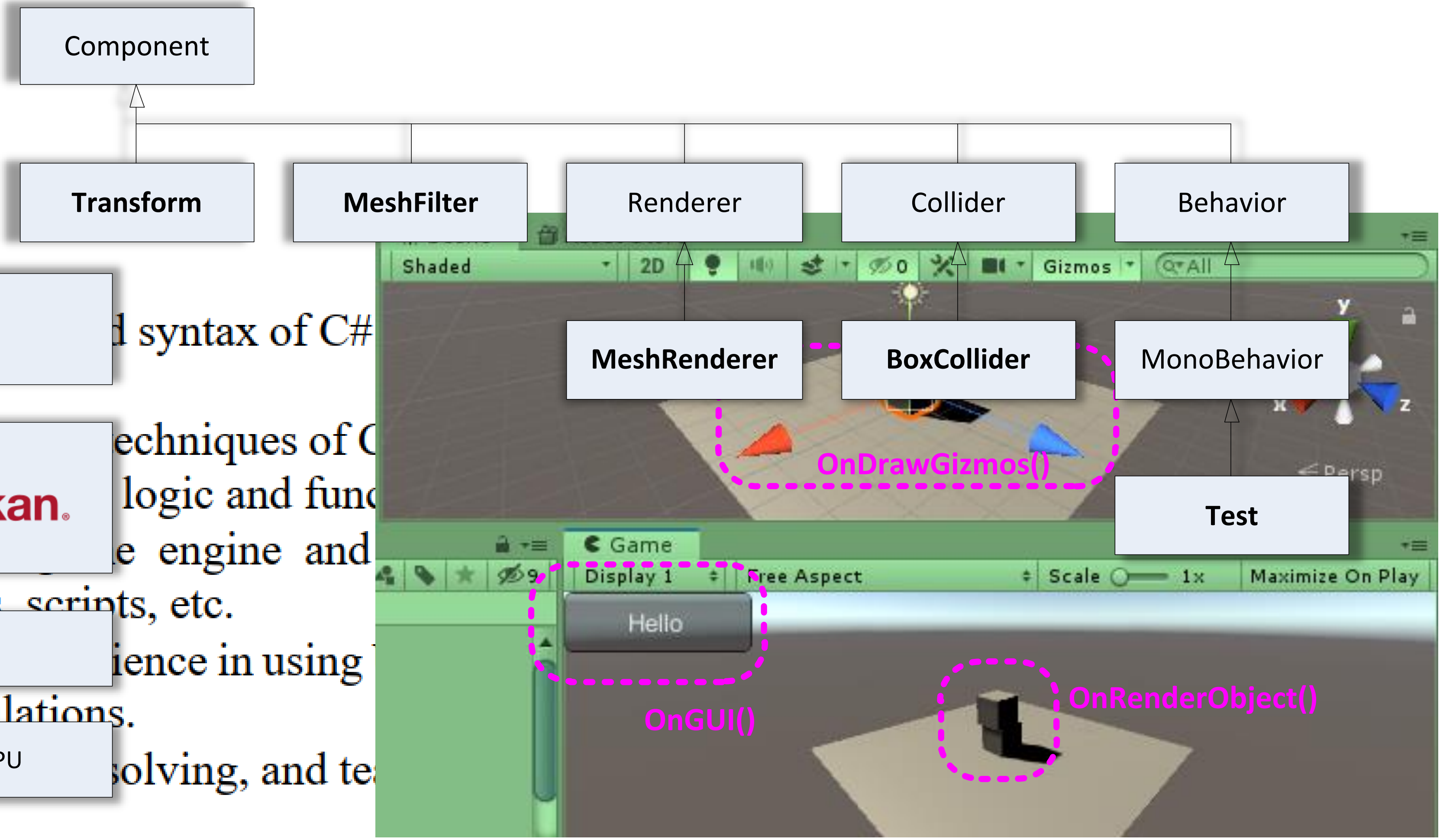
- Understand the basic concepts and principles network architectures, protocols, standards, and models.
- Acquire the skills and knowled de mputer networks using various technologies and tools.
- Explore the current trends and challenges in computer net king, such as security, and cloud computing.
- Develop the ability to communicate eff ely and collaborate with others in projects and tasks.
- Foster the interest and curiosity in computer networking and its applications in various domains.

6th Semester

3D Game Programming – Syllabus

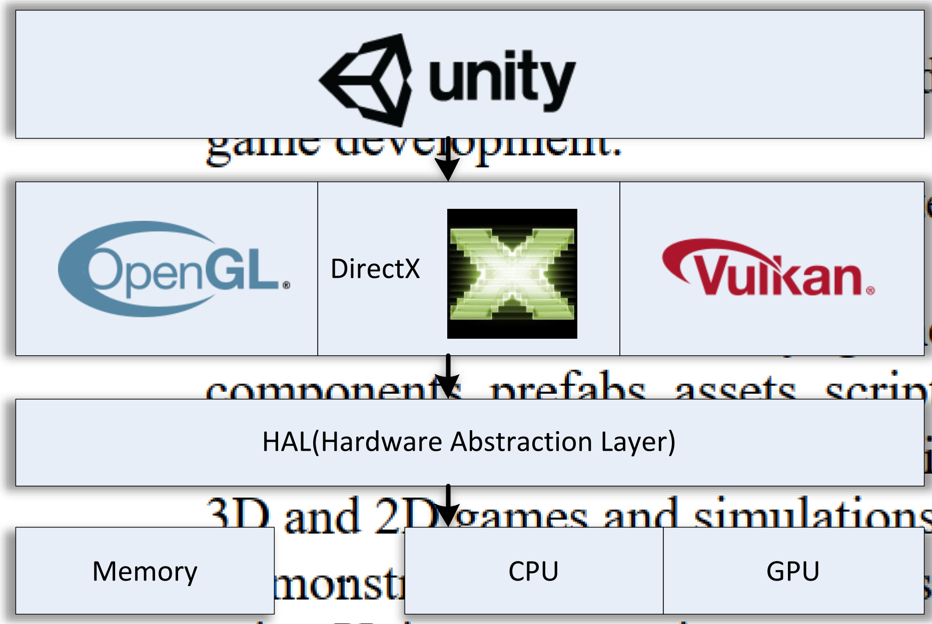
Course Description

The game engine can be used to create three-dimensional (3D) and two-dimensional (2D) games, as well as interactive simulations and other experiences. The engine has been adopted by industries outside video gaming, such as film, automotive, architecture, engineering, construction, and the United States Armed Forces. Students learn Unity game programming.



Goals

The main goals of this course are to:



d syntax of C#
game development.
echniques of C
logic and func
e engine and
components, prefabs, assets, scripts, etc.
ience in using
3D and 2D games and simulations.
monstr
solving, and te
using Unity game engine.

Game Server Programming – Syllabus

Course Description

A game server is a server which is the authoritative source of events in a multiplayer video game. The server transmits enough data about its internal state to allow its connected clients to maintain their own accurate version of the game world for display to players. They also receive and process each player's input. Students learn all techniques used in the game server.

Goals

Let's Share What We Know

The main goals of this course are to:

- ...erst... ba... ...twork communication.
- ...i... ...various models and
...ame servers.
...server prototype.

...rf...

World Wide Web

It is recommended that students take 'Computer Network' courses in the previous semester. If not, students should already know all the basic concepts of networking.

Game Tool Programming – Syllabus

Course Description

C++ was designed with systems programming and embedded, resource-constrained software and large systems in mind, with performance, efficiency, and flexibility of use as its design highlights. We learn how to make game tools with C++.

Goals

The ma

C++17

Windows

es of C++ as a systems programming la

Design Patterns

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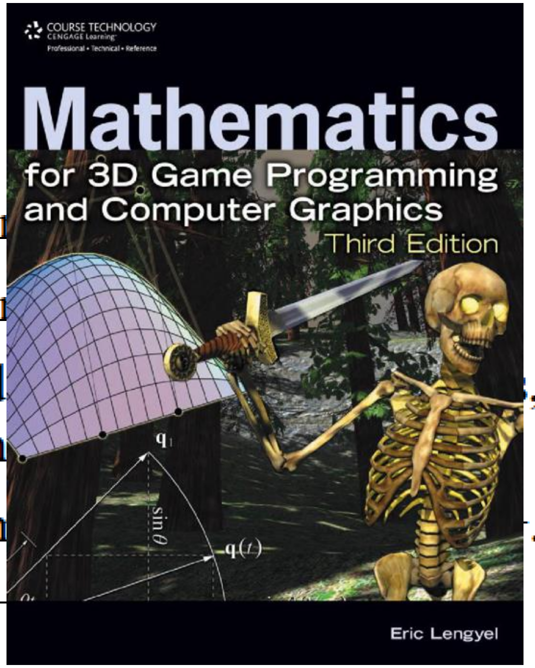
UNREAL ENGINE



Computer Architecture – Syllabus

Course Description

Computer architecture is a description of the structure of a computer system made from component parts. It can sometimes be a high-level description that ignores details of the implementation. This course aims to understand the main concepts of computer architecture for unfamiliar students.

Goals

The main goals of this course are to:

- Introduce the students to the basic concepts and principles of computer architecture, such as data representation, instruction sets, CPU design, memory hierarchy, and peripherals.
- Develop the students' ability to analyze and compare different computer architectures and evaluate their performance and trade-offs.
- Enhance the students' skills in programming and logic circuits using various tools and languages.
- Provide the students with hands-on experience in designing and implementing simple computer architectures using simulation software and hardware components.
- Prepare the students for further studies and research in computer engineering and related fields.

Operating System – Syllabus

Course Description

Operating systems are an essential part of any computer system. Similarly, a course on operating systems is an essential part of any computer-science education. The design of an operating system is an important factor in developing good software. So, we must learn the basic principles of designing the operating system to develop efficient software.

Goals

The main goals of this course are to:

- Understand the basic concepts and principles of operating systems design and implementation.
- Compare and contrast different types of operating systems, such as batch, interactive, multiprogrammed, distributed, and real-time systems.
- Analyze and evaluate various process management techniques, such as process creation, scheduling, synchronization, communication, and deadlock handling.
- Apply the concepts of multithreading and concurrency to develop parallel and distributed applications.
- Explain and demonstrate how memory management and virtual memory work in modern operating systems.

7th Semester

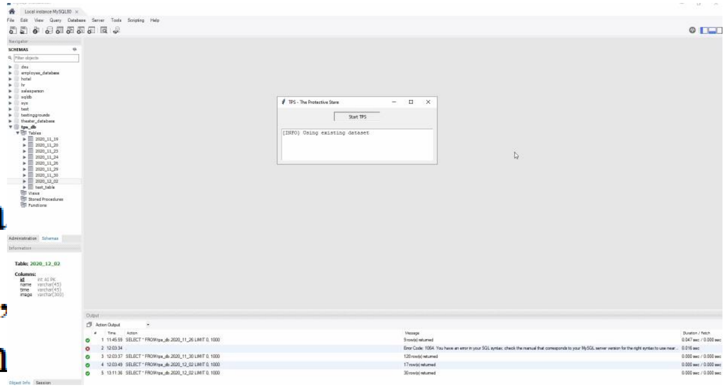
Bachelor Thesis 1 – Syllabus

Course Description

Students must do a graduation project for graduation. Bachelor Thesis classes are arranged for the graduation project. You need to write a game design document and prepare the prototype version of the game.

Goals

The main goals of this course are to:



- Develop the skills and knowledge nec
- Apply the principles and techniques internal economy, machinations, desi mechanics to the game project.
- Demonstrate the ability to write a clea scope, features, mechanics, systems, a
- Produce a playable prototype of the g
- Evaluate the strengths and weaknesse and testing.

Research Methods – Syllabus

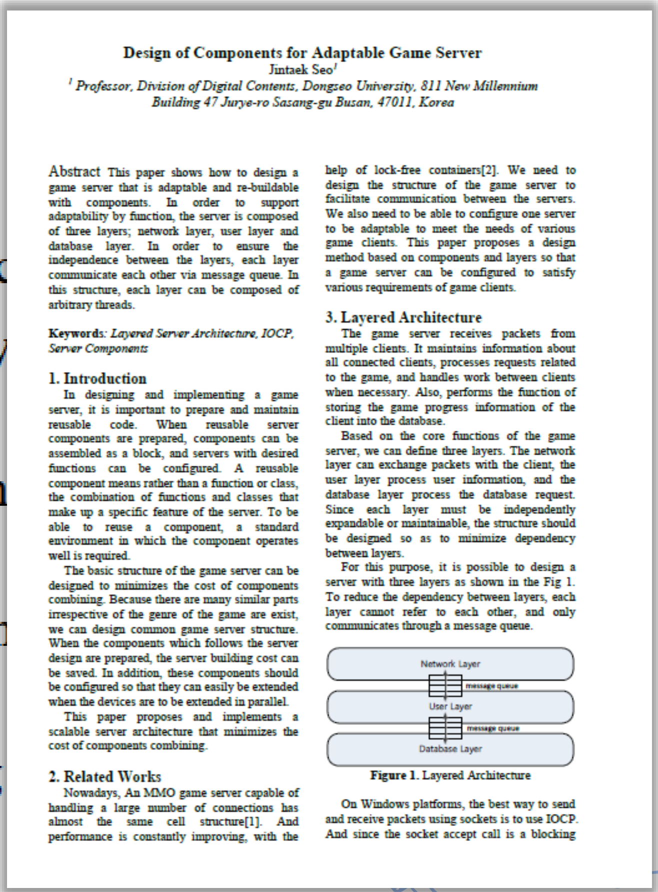
Course Description

Learn to analyze the issues /goals/objectives/tasks raised in order to find the necessary information to analyze the decisions, assessments, and practical proposals for experimental research work.

Goals

The main goals of this course are to:

- Develop the skills and knowledge necessary to conduct and evaluate researc
- Understand the principles and methods of scientific inquiry and how they research questions and designs.
- Know how to critically read, analyze, and synthesize research papers an findings effectively in written and oral forms.
- Design, implement, and report on a research project that addresses a relevan chosen field of study.
- Demonstrate ethical and professional standards in conducting and reporting



Design of Components for Adaptable Game Server

Jintaek Seo[1]

[1] *Professor, Division of Digital Contents, Dongseo University, 811 New Millennium Building 47 Jurye-ro Sasang-gu Busan, 47011, Korea*

Abstract This paper shows how to design a game server that is adaptable and re-buildable with components. In order to support adaptability by function, the server is composed of three layers; network layer, user layer and database layer. In order to ensure the independence between the layers, each layer communicate each other via message queue. In this structure, each layer can be composed of arbitrary threads.

Keywords: *Layered Server Architecture, IOCP, Server Components*

1. Introduction

In designing and implementing a game server, it is important to prepare and maintain reusable code. When reusable server components are prepared, components can be assembled as a block, and servers with desired functions can be configured. A reusable component means rather than a function or class, the combination of functions and classes that make up a specific feature of the server. To be able to reuse a component, a standard environment in which the component operates well is required.

The basic structure of the game server can be designed to minimizes the cost of components combining. Because there are many similar parts irrespective of the genre of the game are exist, we can design common game server structure. When the components which follows the server design are prepared, the server building cost can be saved. In addition, these components should be configured so that they can easily be extended when the devices are to be extended in parallel.

This paper proposes and implements a scalable server architecture that minimizes the cost of components combining.

2. Related Works

Nowadays, An MMO game server capable of handling a large number of connections has almost the same cell structure[1]. And performance is constantly improving, with the help of lock-free containers[2]. We need to design the structure of the game server to facilitate communication between the servers. We also need to be able to configure one server to be adaptable to meet the needs of various game clients. This paper proposes a design method based on components and layers so that a game server can be configured to satisfy various requirements of game clients.

3. Layered Architecture

The game server receives packets from multiple clients. It maintains information about all connected clients, processes requests related to the game, and handles work between clients when necessary. Also, performs the function of storing the game progress information of the client into the database.

Based on the core functions of the game server, we can define three layers. The network layer can exchange packets with the client, the user layer process user information, and the database layer process the database request. Since each layer must be independently expandable or maintainable, the structure should be designed so as to minimize dependency between layers.

For this purpose, it is possible to design a server with three layers as shown in the Fig 1. To reduce the dependency between layers, each layer cannot refer to each other, and only communicates through a message queue.

Figure 1. Layered Architecture

On Windows platforms, the best way to send and receive packets using sockets is to use IOCP. And since the socket accept call is a blocking

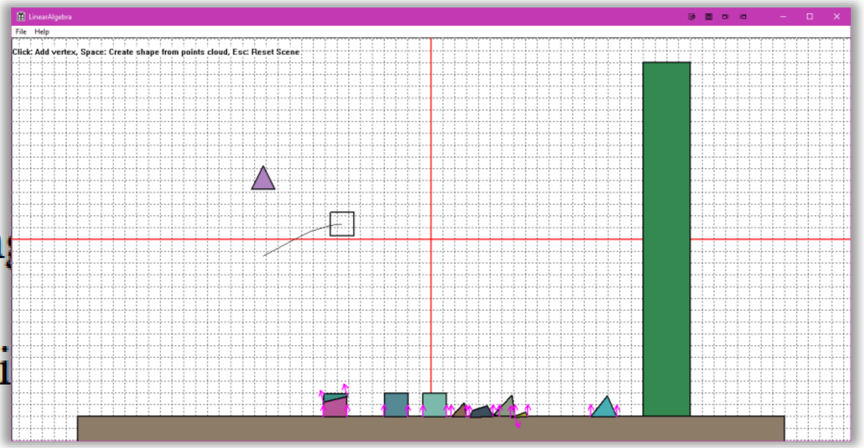
Using Engine Game – Syllabus

Course Description

This course provides a comprehensive introduction to the implementation of game physics engines. It is designed for those who are interested in understanding the underlying principles and mechanics of game physics and how they are applied in modern game development. Key topics covered in the course include: Fundamental Mathematics, Physics Concepts, Rigid Body Dynamics, Physics Engine Development, Constraints.

Goals

The main goals of this course are:



- Understand Fundamental Concepts: Gain a solid understandin principles of game physics.
- Develop Practical Skills: Develop practical skills in implementi development.
- Apply Mathematical Concepts: Apply mathematical concepts to solve problems related to game physics.
- Analyze and Design: Analyze and design game physics engines for various types of games.
- Explore Advanced Topics: Explore advanced topics in game physics, such as fluid and cloth simulation.

Game Project – Syllabus

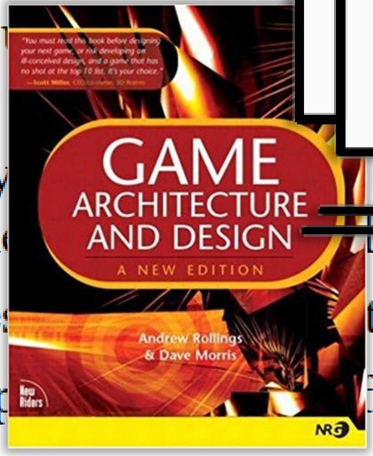
Course Description

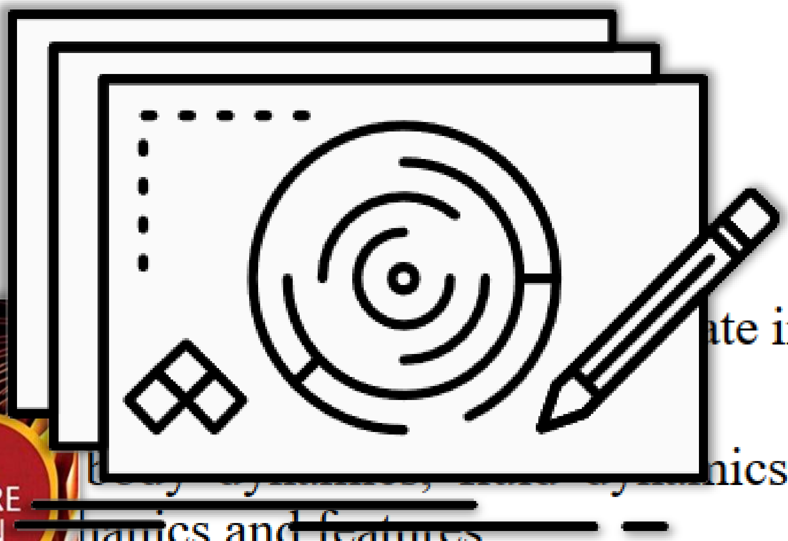
This course provides an introduction to Extended Reality (XR), a term that encompasses Virtual Reality (VR), Augmented Reality (AR), and Mixed Reality (MR). The course focuses on how the underlying technologies of XR have evolved over time and launched the concepts of VR and AR into mainstream consciousness. This course also introduces you to Unity Machine Learning Agents (ML-Agents), a toolkit that allows you to create more compelling gameplay and an enhanced game experience. Students will learn how to train intelligent agents using a combination of deep reinforcement learning and imitation learning.

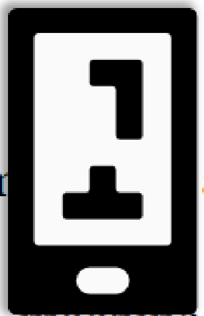
Goals

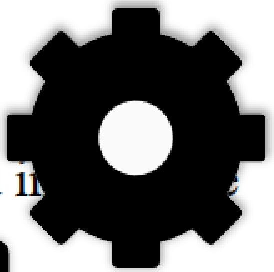
The main goals of this course are to:

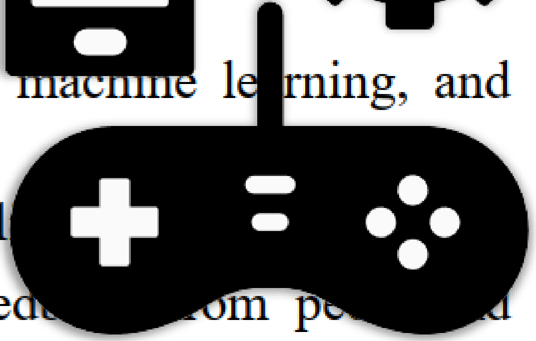
- Gain practical experience in … ate in … and i … VR games.
- Apply the concepts of play … ics, machine learning, and inference to design and impl … anics and features.
- Demonstrate the ability to te … timize game performance and qual …
- Showcase the final game p … fessional manner and receive feed … om pe … d instructors.











8th Semester

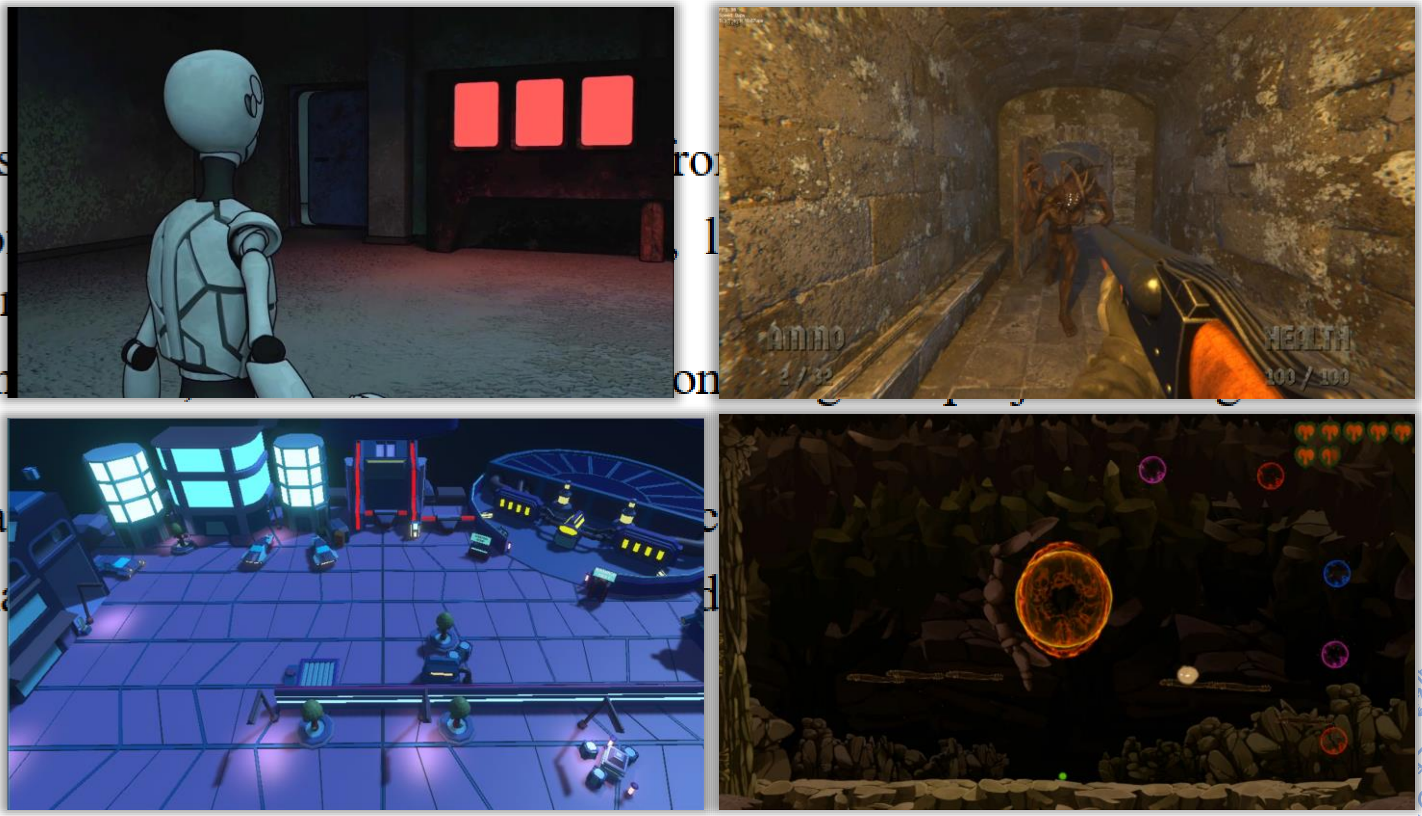
Bachelor Thesis 2 – Syllabus

Course Description

Students must do a graduation project for graduation. Bachelor Thesis classes are arranged for the graduation project. You need to write a game design document and prepare the final version of the game.

Goals

The main goals of this course are to:



- Develop the skills and knowledge necess
- Apply the principles and techniques o optimization to create a functional and e
- Demonstrate the ability to present, comm the development process.
- Polish a game design document and a ga
- Reflect on the learning outcomes and ch

Prerequisites (optional)

Students must take the 'Bachelor Thesis 1' course offered in the previous semester.

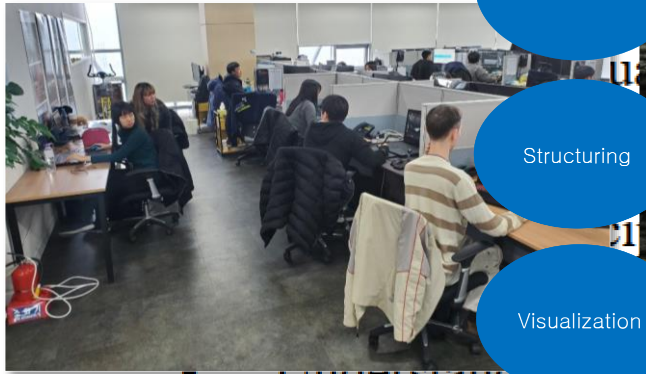
Portfolio – Syllabus

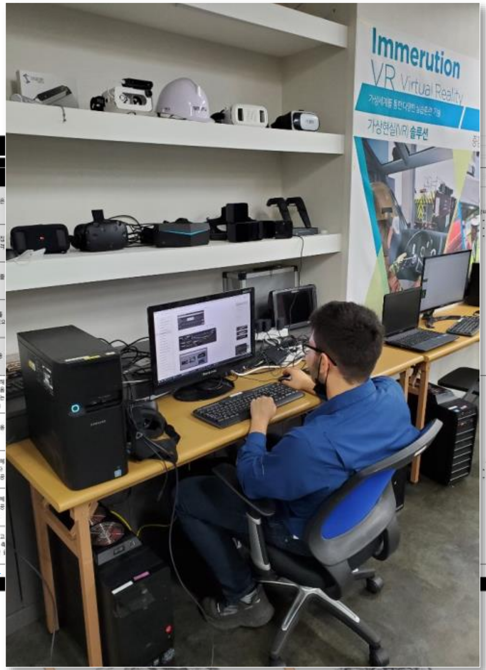
Course Description

This subject is designed to help students prepare effective resumes and portfolios that showcase their skills and competencies in the IT field. The course covers various types of resumes and portfolios, as well as the tools and techniques to create and manage them. The course also introduces the concepts and practices of IT portfolio management, project portfolio management, and enterprise project portfolio management.

Goals

The main goals of thi





- Understand
- Recognize and a organizational pe

Sound Workshop – Syllabus

Course Description

Beyond the basic recording and editing of sound, you experience a more active and creative sound design. In addition, students learn techniques to create and express various sounds that fill the artistry and completeness of the film. In addition, students learn skills to compose simple music by using various music software.

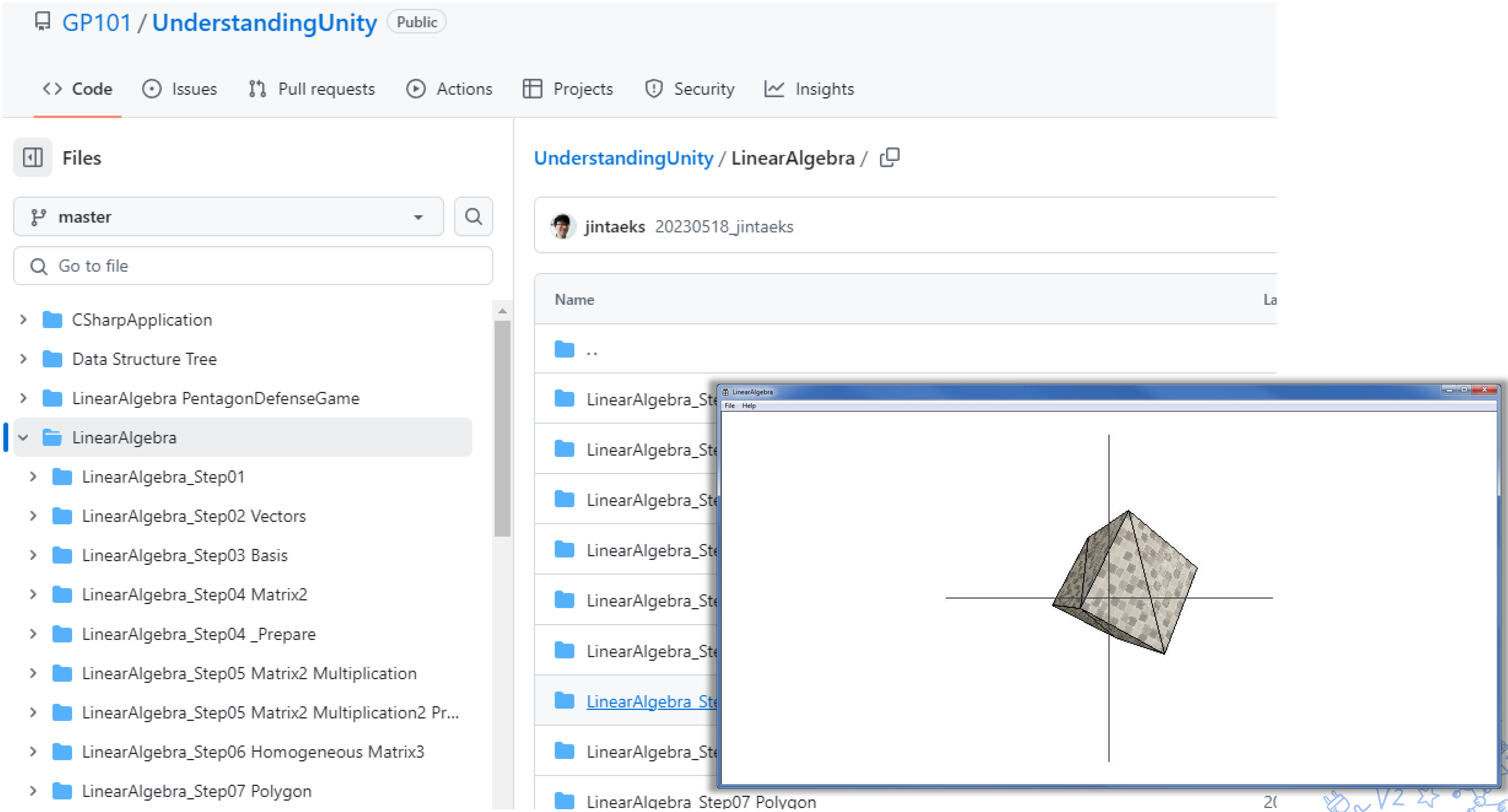
Goals

The main goals of this course are to:

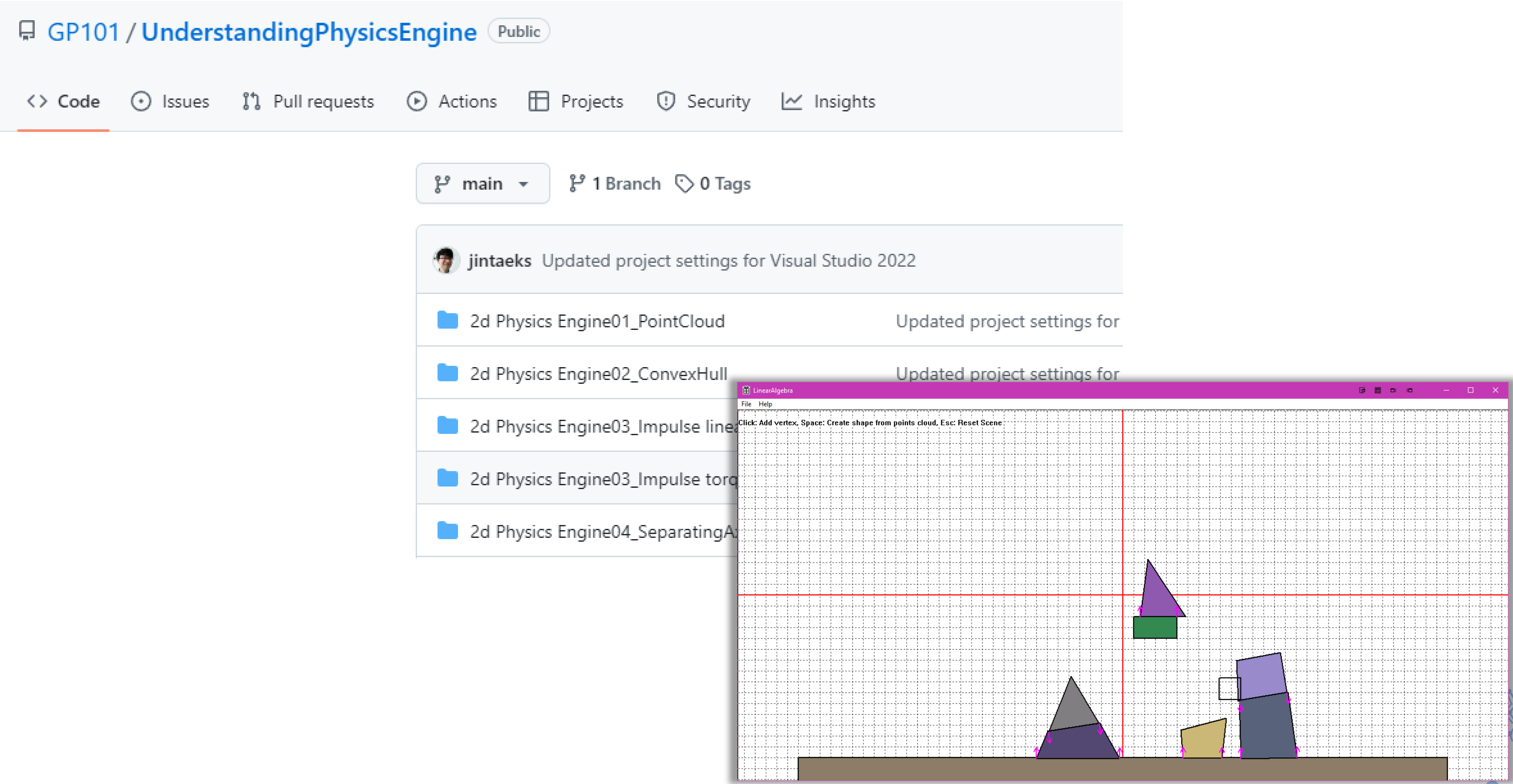
- Understand the principles and techniques of professional audio post production for film and other media.
- Develop creative and expressive sound design skills that enhance the artistic and emotional impact of the film.
- Analyze and critique the use of sound and music in various film genres and styles.
- Learn how to use various audio software and tools to create, edit, process and mix sound and music.
- Compose simple but effective musical scores that match the mood, theme and genre of the film.

Open Source Repository

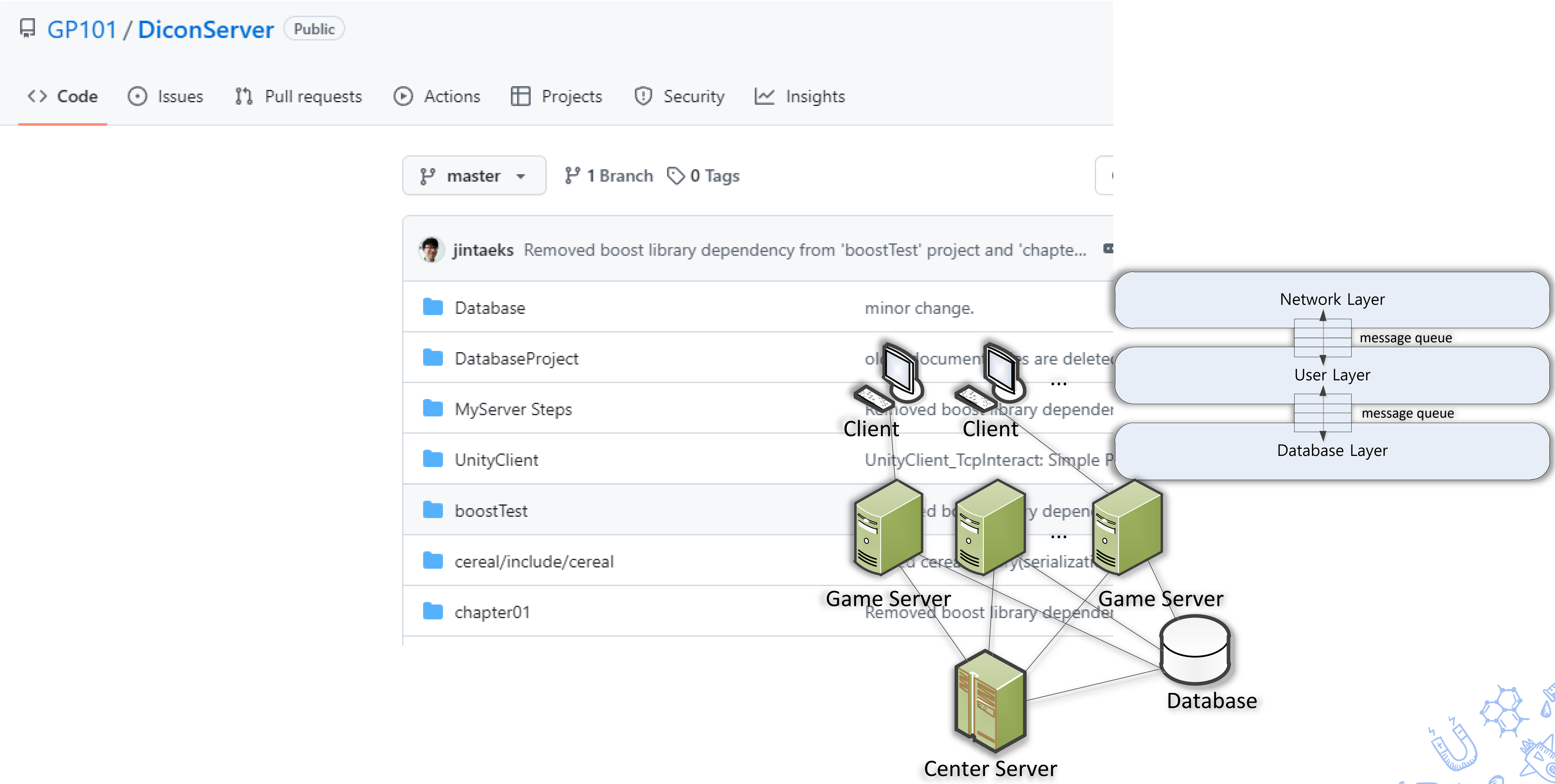
GP101: Rendering Engine



GP101: Physics Engine



GP101: Game Server



Graduation Regulation

3. Preparatory procedures for graduation application

3-1. Details of the preliminary review of the qualification for applying for graduation work

① Graduation candidates must receive an evaluation of their portfolio and graduation works.

② Portfolio: The supervisor determines the case in which one's major exists within the relevant industry group and approval.

③ Graduation work

a. Proposal : summary of graduation project or game design, etc., submitted in paper, and presented.

b. Final result : Final Graduation Work, Presentation file for final Graduation Work, Final Demo Video and Executable file.

3-2. Exceptions

① Students who are unable to participate in the production of graduation works due to reasons recognized by the graduation assessment committee may qualify for graduation after satisfying the conditions of the works and thesis set by the committee.

② In the event of exceptional circumstances (public contest, thesis, etc.) that are not specified in this document, they may be eligible for graduation through deliberation by the Graduation assessment Committee.

4. Method and contents of application for graduation assessment

4-1. Prospective graduates must apply for graduation assessment at least one semester prior to the scheduled date of graduation in accordance with the provisions specified below.

① Prepared prospective graduates shall apply for and receive evaluations, First ⇨ Final etc., according to the specified contents below.

② The application for graduation review will be completed by submitting the materials required by the graduation assessment committee to the International College.

4-2. Submissions and review stage

① First assessment

a. Application : Study Plan of Graduation Work : PDF

b. Portfolio : Resume, Structure, Demo Video : PDF

c. Graduation Work

-Proposal : The Proposal of Graduation Work : PDF

- ICT Project: Draft thesis, Graduation Work Presentation File, Exe File and Demo Video.
- Game Project : Draft Game Design Document, Graduation Work Presentation File, Exe File and Demo Video.

※ The submitted materials that doesn't meet the specification will not be judged.

Facilities

Creative Arts Studio

| Creative Arts Studios : A learning ecosystem based on the production pipeline, including ChromaKey, Sound Recording, and Animation Studio, is located at Creative Arts Studios in the University-Industry Foundation.

Sound Recoding Studio
-Design, mixing, foley(Prof.김민주)
•

Faculty Office 1
-(Prof.)
•

Project Incubator
-(Prof.)
•

Chroma key Studio
-Visual Effects (Kim, Kyoungheon)
-Lighting Creation (Han, Yooneong)
-Matte Painting (Ji, Yun)

• Necessary for improving video images such as special effects, lighting, compositing research space.
• Linked Research Group(storytelling, modeling, animation)

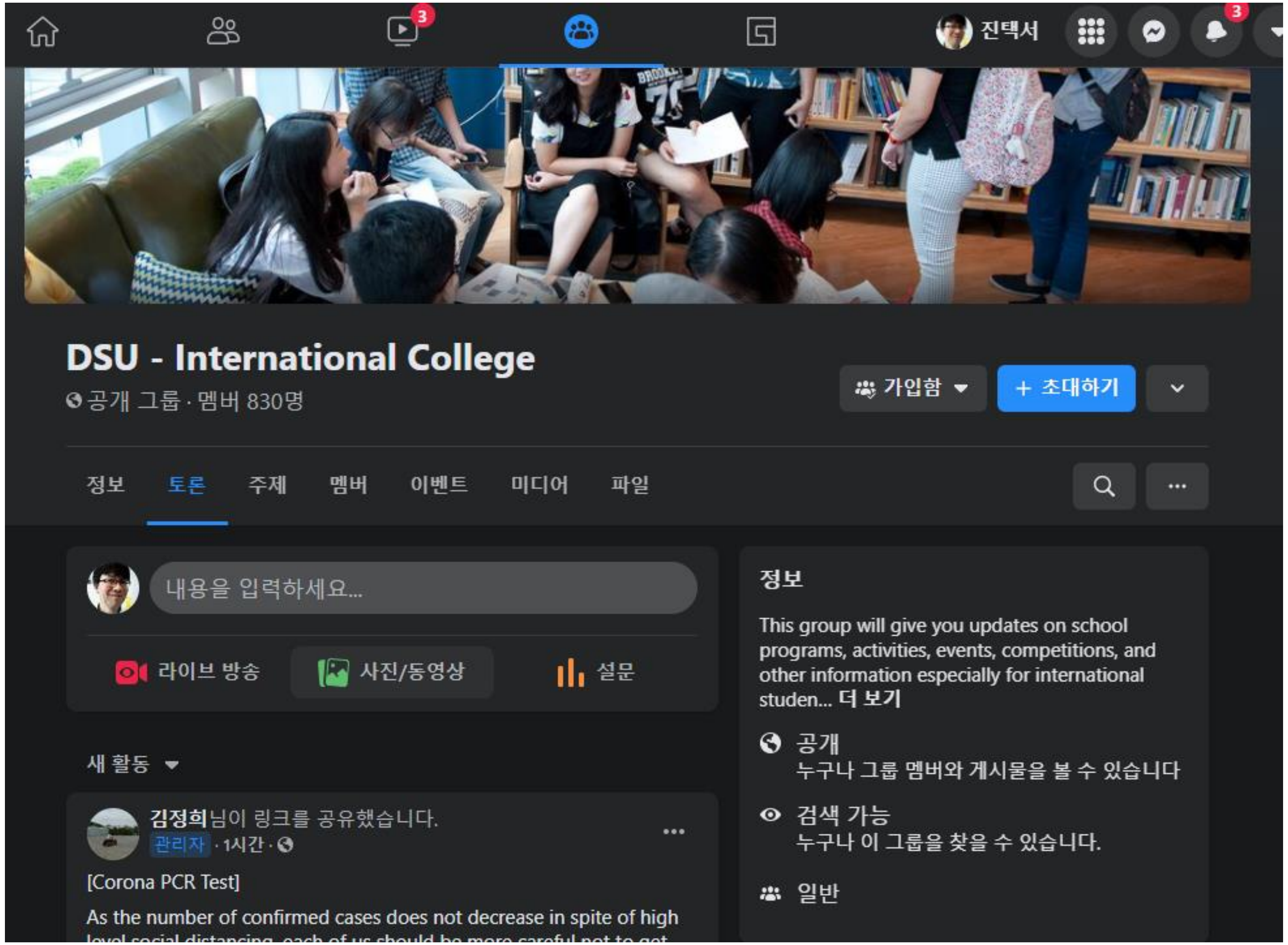
Faculty Office 2
-(Prof.)
•

Animation & Motion Capture Studio
-Animation Studio(이재호, Qu, Lin)

• It is the lab for performing research on animation theory and practical skills and solve problems that occur during project production. Linked Research Group (Story Telling & Character Setup)

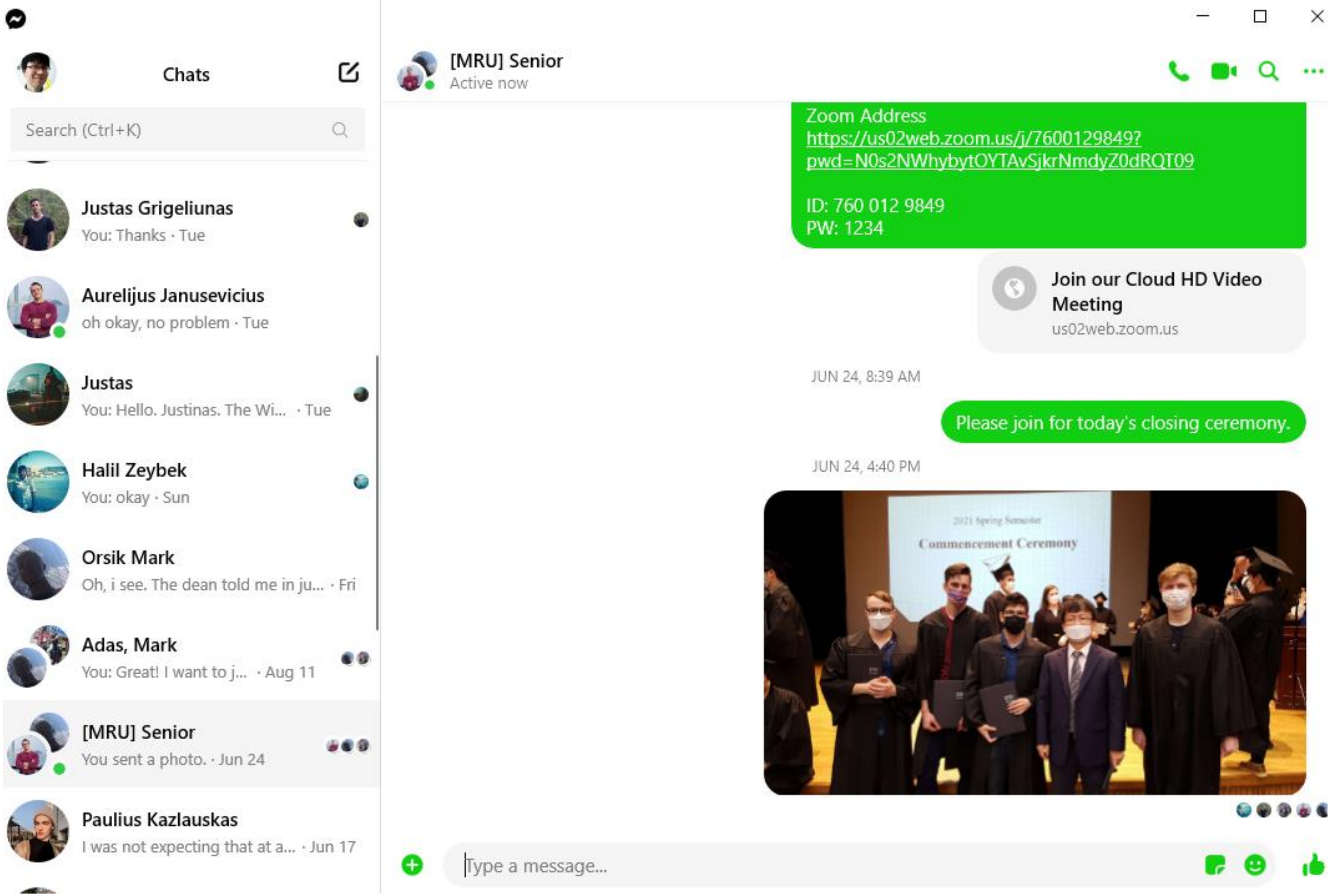
◆ Facebook DSU IC Group

https://www.facebook.com/groups/1718863521704881



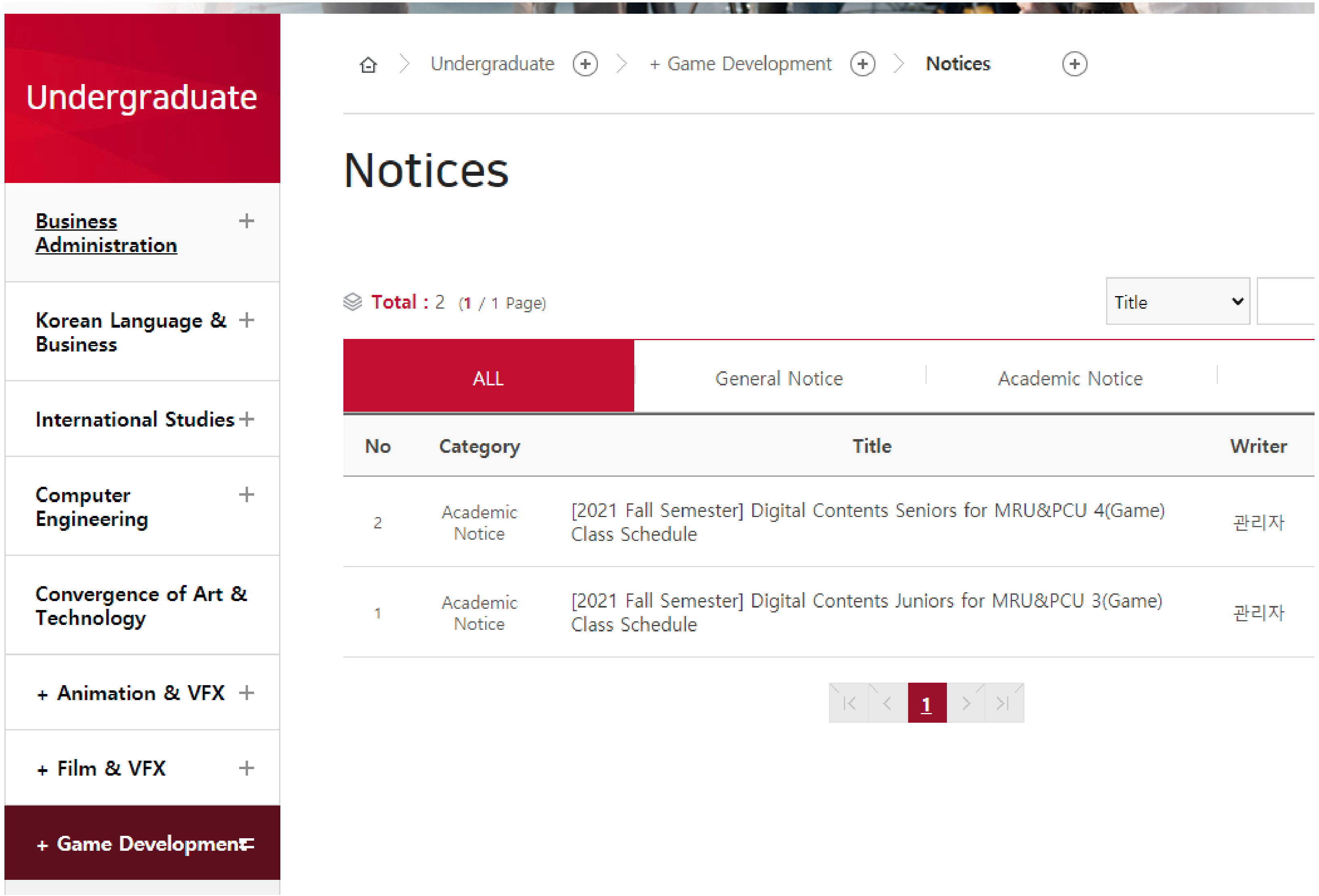
Facebook Messenger

https://www.facebook.com/Jintaek.Seo88/



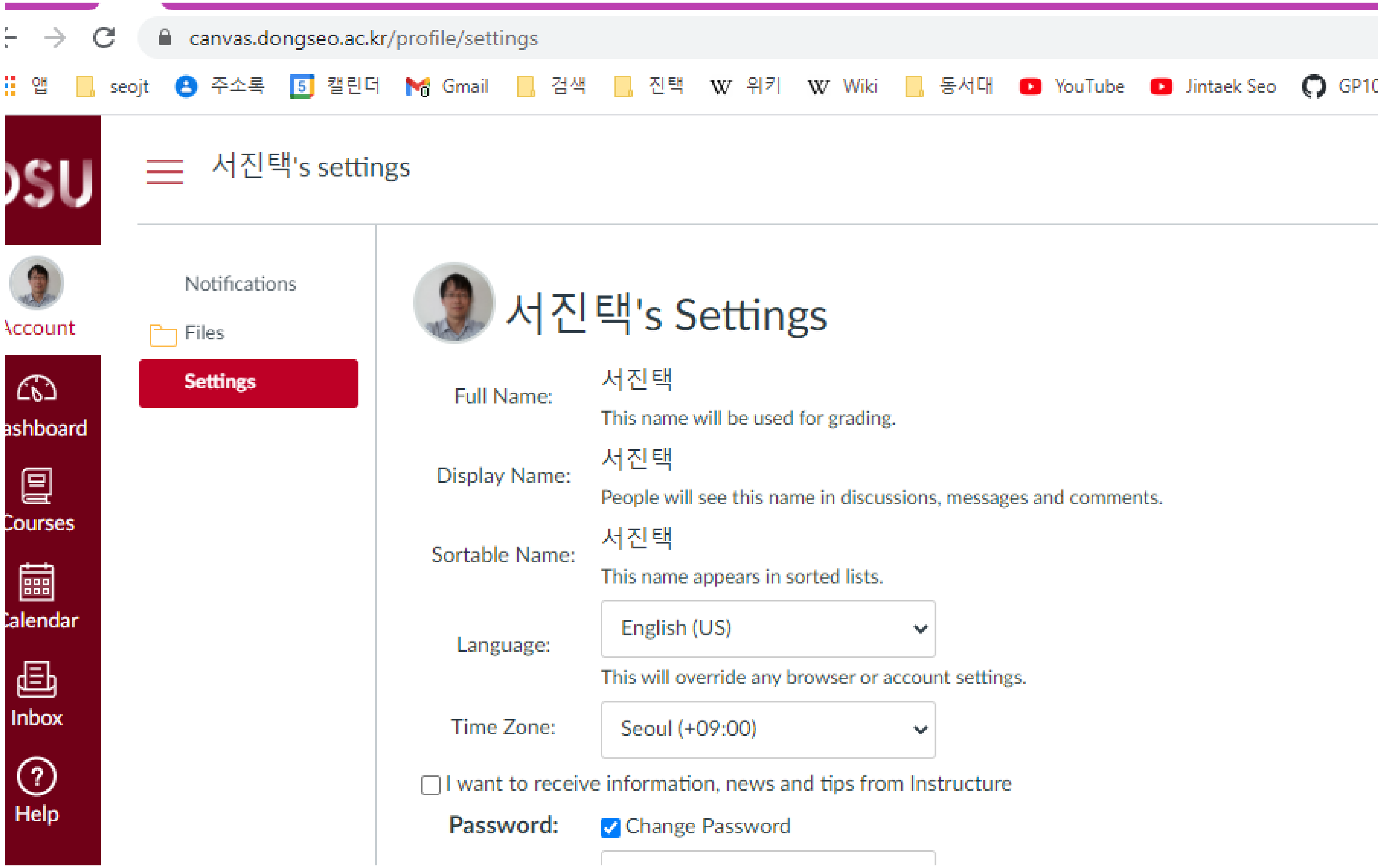
DSU IC Homepage

https://uni.dongseo.ac.kr/ic/index.php?pCode=



LMS

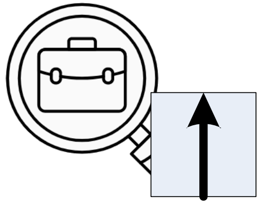
https://eclass1.dongseo.ac.kr



How can I find my dream?

How can I find my dream?

A



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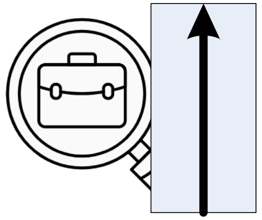

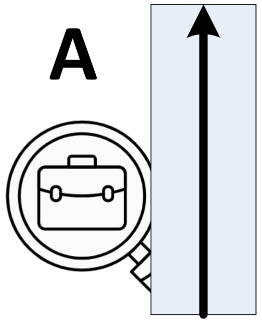
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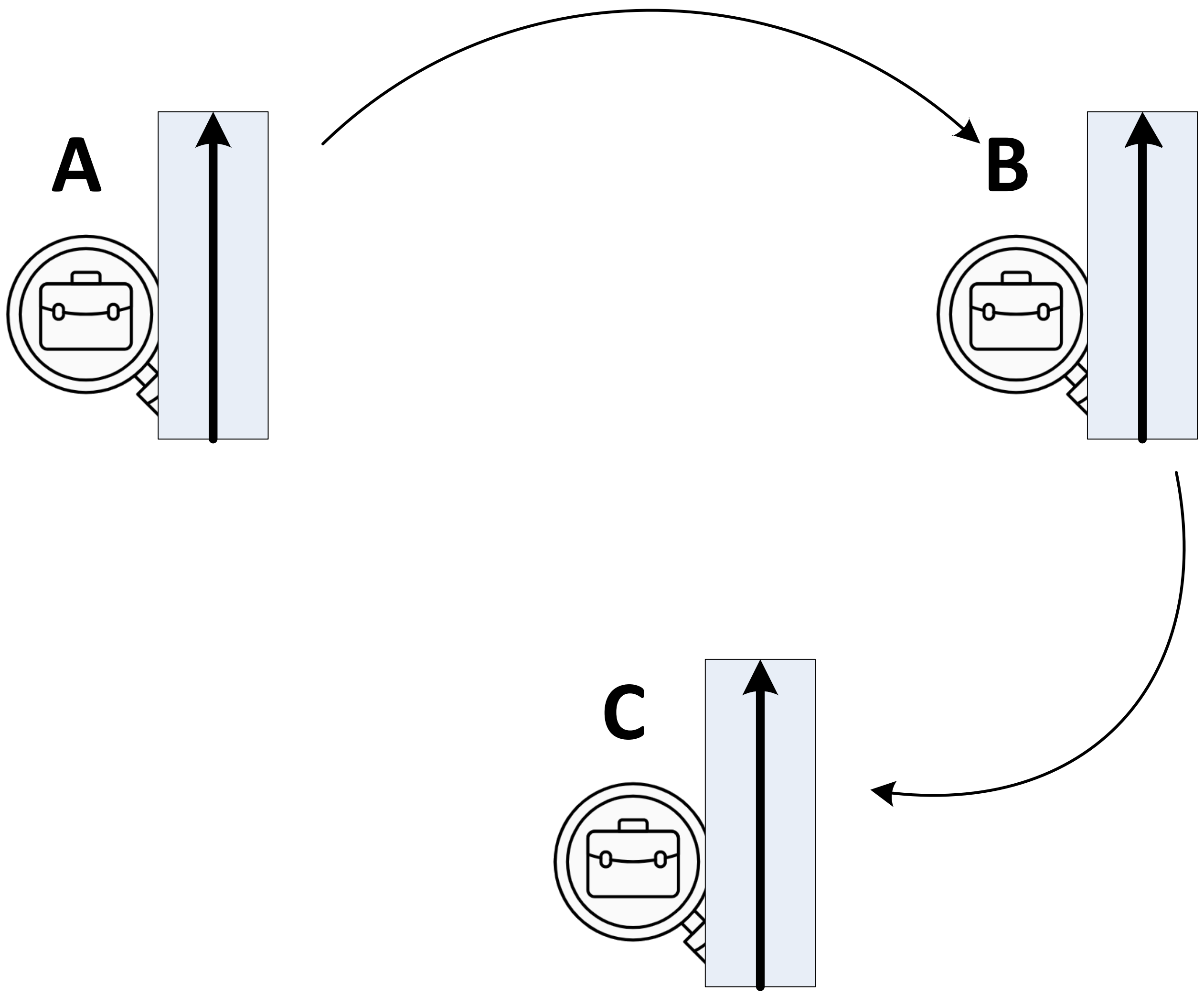
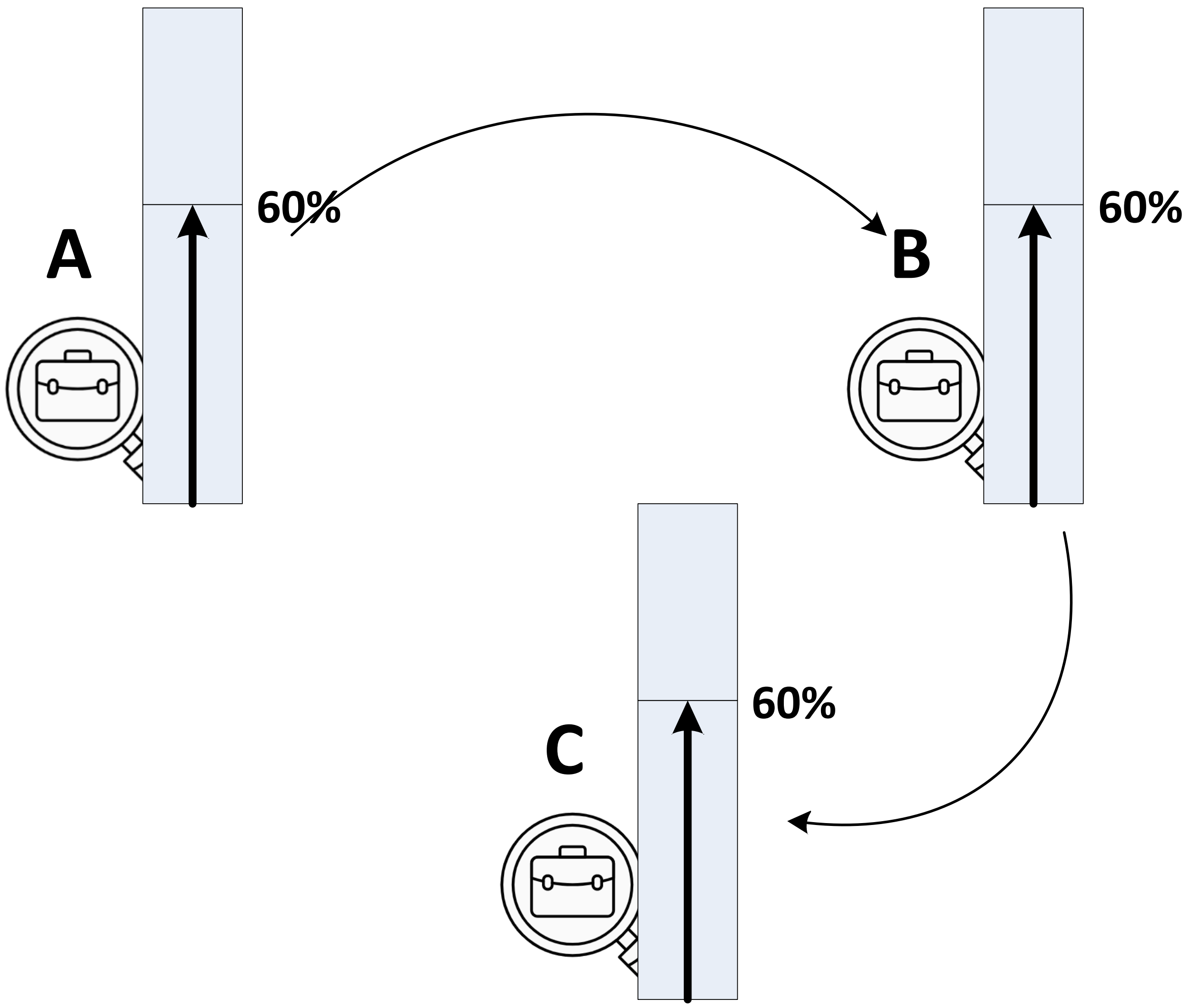

A
B
C


60%
A
60%
B
60%
C

How can I find my dream?

✓ Do NOT do what you like.

✓ Like what you do.

MY **BRIGHT** FUTURE

DSU **Dongseo** University
동서대학교